

AI POWERED LOAN APPROVAL ANALYSIS SYSTEM

n8n Agentic Machine Learning Pipeline

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PROJECT OVERVIEW:

This project is a fully automated, agentic machine learning pipeline built using n8n Cloud, FastAPI, Python, and GPT-4o. The system enables a bank analyst to upload a loan applicant dataset and receive a comprehensive AI-generated risk assessment report – with zero manual intervention beyond the initial upload.

- Workflow 1 - Portfolio Analysis: Upload a CSV of loan applicants and receive a full ML-powered portfolio risk report as a PDF.
- Workflow 2 – Individual Loan Lookup: Enter a Loan ID and receive a personalized AI-written decision letter explaining the approval or rejection.

BUSINESS PROBLEM:

Traditional loan approval processes are manual, time-consuming, and inconsistent. Banks process thousands of applications requiring human review of financial profiles lending criteria. The project demonstrates how an agentic AI pipeline can automate the full analytical cycle – from raw data ingestion to explainable, audit-ready decisions – reducing processing time from hours to minutes while maintaining transparency and compliance.

TECHNOLOGIES USED:

Orchestration & Automation:

The workflow orchestration layer is built on n8n cloud, which handles the full agentic pipeline – from form – based CSV ingestion through multi-step HTTP requests, OpenAI API calls, and final report delivery. Since n8n Cloud cannot directly access local machine resources, ngrok is used to create a secure public tunnel from n8n Cloud to the locally hosted FastAPI server, enabling seamless communication between the cloud orchestrator and the on-premise ML execution environment.

Machine Learning:

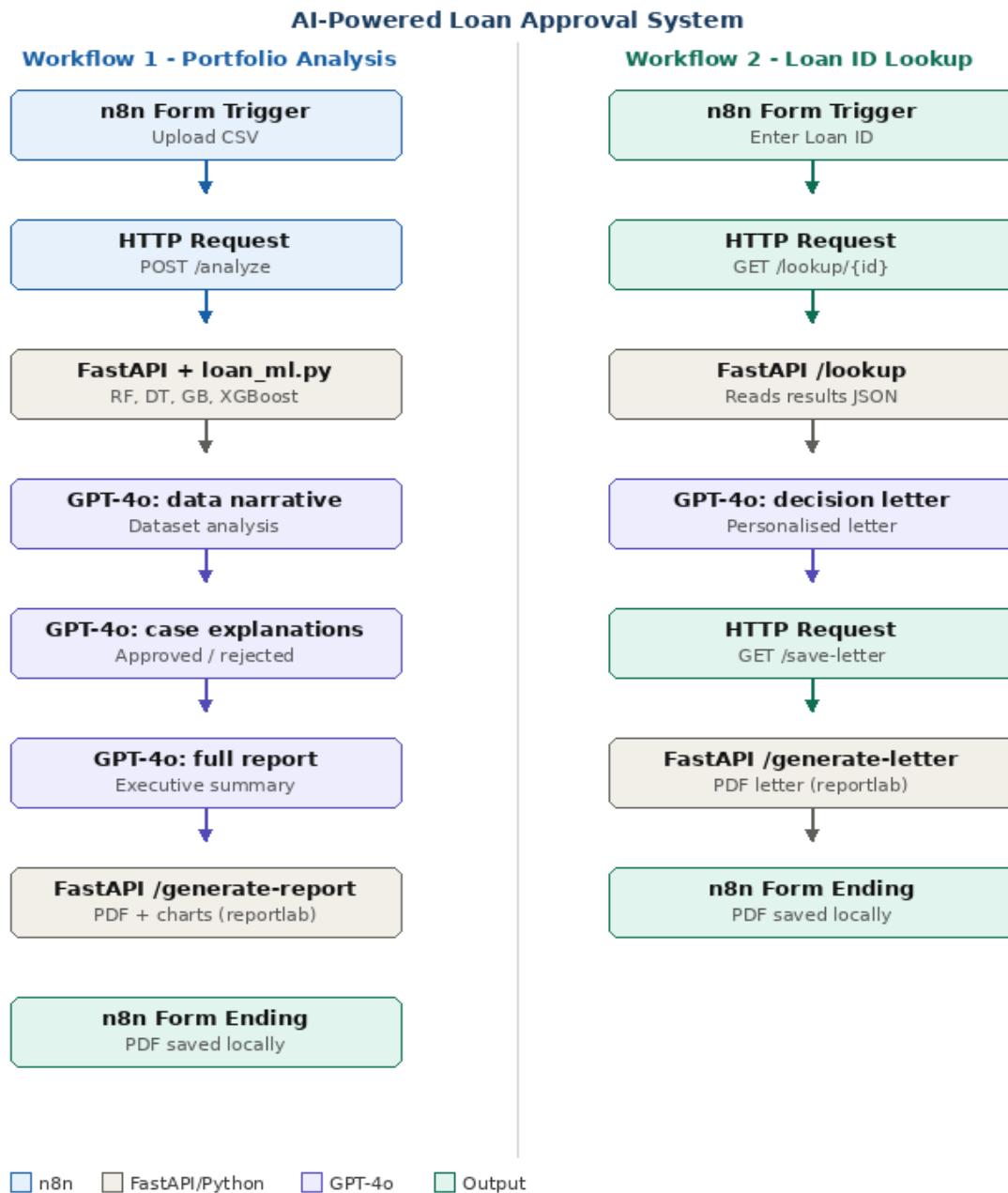
The analytical core is written in Python, leveraging the scikit-learn library to train and evaluate three classification models – Random Forest, Decision Tree, and Gradient Boosting - alongside XGBoost as a fourth high-performance boosting alternative. Data ingestion, cleaning, and feature engineering are handled by pandas and NumPy, which are used to construct four ratio-based engineered features from the raw applicant data. Model performance is visualised using matplotlib and seaborn, producing feature importance charts and model comparison bar charts that are embedded directly into the final PDF report.

API & Report Generation:

FastAPI serves as the REST API microservice layer, exposing five endpoints that bridge n8n's workflow nodes to the local Python environment. The server is run using uvicorn, a high-performance ASGI server suited to asynchronous FastAPI applications. PDF generation is handled by reportlab, which assembles the structured report text and embedded chart images into a professionally formatted A4 document. OpenAI's GPT-4o model is integrated at three points in the pipeline - generating a data understanding narrative, writing individual case explanations, and authoring personalised applicant decision letters - ensuring that all outputs are human-readable, contextually accurate, and audit-ready.

SYSTEM ARCHITECTURE:

The diagram below illustrates both workflows, showing how n8n Cloud, FastAPI, the Python ML layer, and GPT-4o connect end to end.



KEY RESULTS:

- Best Model: Decision Tree (ROC-AUC: 0.6073, F1: 0.2715, Accuracy: 62.3%)
- Top Risk Factors: Loan-to-Income Ratio (0.2265), Loan Term (0.1968), Loan-to-Assets Ratio (0.0936)
- Portfolio Approval Rate: 85.48%
- Risk Distribution: 2,866 Medium Risk, 783 Low Risk, 620 High Risk
- Note: CIBIL score intentionally excluded from features to demonstrate model performance on financial ratios alone, improving explainability and fairness.

PROJECT STRUCTURE:

Loan Approval - n8n/

```
|— loan_ml.py           # ML pipeline script
|— api.py              # FastAPI server (5 endpoints)
|— loan_approval_dataset.csv # Input dataset
|— loan_results.json    # ML output (auto-generated)
|— loan_report.pdf      # Portfolio analysis report
|— feature_importance.png # Chart (auto-generated)
|— model_comparison.png  # Chart (auto-generated)
```

PORTFOLIO RELEVANCE:

- End-to-end ML pipeline design and deployment (data → model → API → report)
- Agentic AI workflow orchestration using n8n with multi-step reasoning
- Explainable AI – feature importance, personalised decision reasoning, risk tiering
- REST API development with FastAPI for ML microservice architecture
- Real-world business framing – credit risk, compliance, audit-ready outputs
- GPT-4o integration for automated narrative generation and personalised communications.